Lecture 2.1: Bond Prices, Yields, and Arbitrage

Bond Prices, Yields, and Arbitrage

- Specifically, we're going to study the yield curve
  1. What the yield curve can tell us about the economy
  2. What current yields tell us about future yields
  3. How to price bonds using the yield curve
- We're going to focus on US Treasuries because unlike corporate bonds or government bonds from the developing world, US Treasuries are considered to be risk-free
- We will use the yield curve to compute
  1. zero rates,
  2. forward rates, and
  3. yield-to-maturity
- We will use the yield curve to discount cash flows

Treasury Bills, Notes, and Bonds

- Most marketable of all money market instruments
- Terms:
  - Bills: 1, 2, 3, 6, 12 months
  - Notes: 2, 3, 5, 7, 10 years
  - Bonds: 20, 30 years
- Notes and bonds pay coupons, bills do not (zero-coupon)
- Treasuries are highly liquid
- The income earned on Treasuries is taxable at the federal level, it is exempt from state and local taxes

Treasuries

A T-year Treasury makes semi-annual payments.

0	1	2	$\cdots$	$2T$
$-P$	$cF/2$	$cF/2$	$\cdots$	$cF/2 + F$

where $c \geq 0$ is the coupon rate (APR)

$$P \overset{\text{should}}{=} \frac{cF/2}{(1+y/2)} + \frac{cF/2}{(1+y/2)^2} + \frac{cF/2}{(1+y/2)^3} + \cdots + \frac{cF/2}{(1+y/2)^{2T}} + \frac{F}{(1+y/2)^{2T}}$$
$$= \frac{cF/2}{y/2}\left(1 - \frac{1}{(1+y/2)^{2T}}\right) + \frac{F}{(1+y/2)^{2T}}$$

where y is the yield (APR).

TREASURY NEWS

Department of the Treasury • Bureau of the Fiscal Service

For Immediate Release
August 13, 2020

CONTACT: Treasury Auctions
202-504-3550

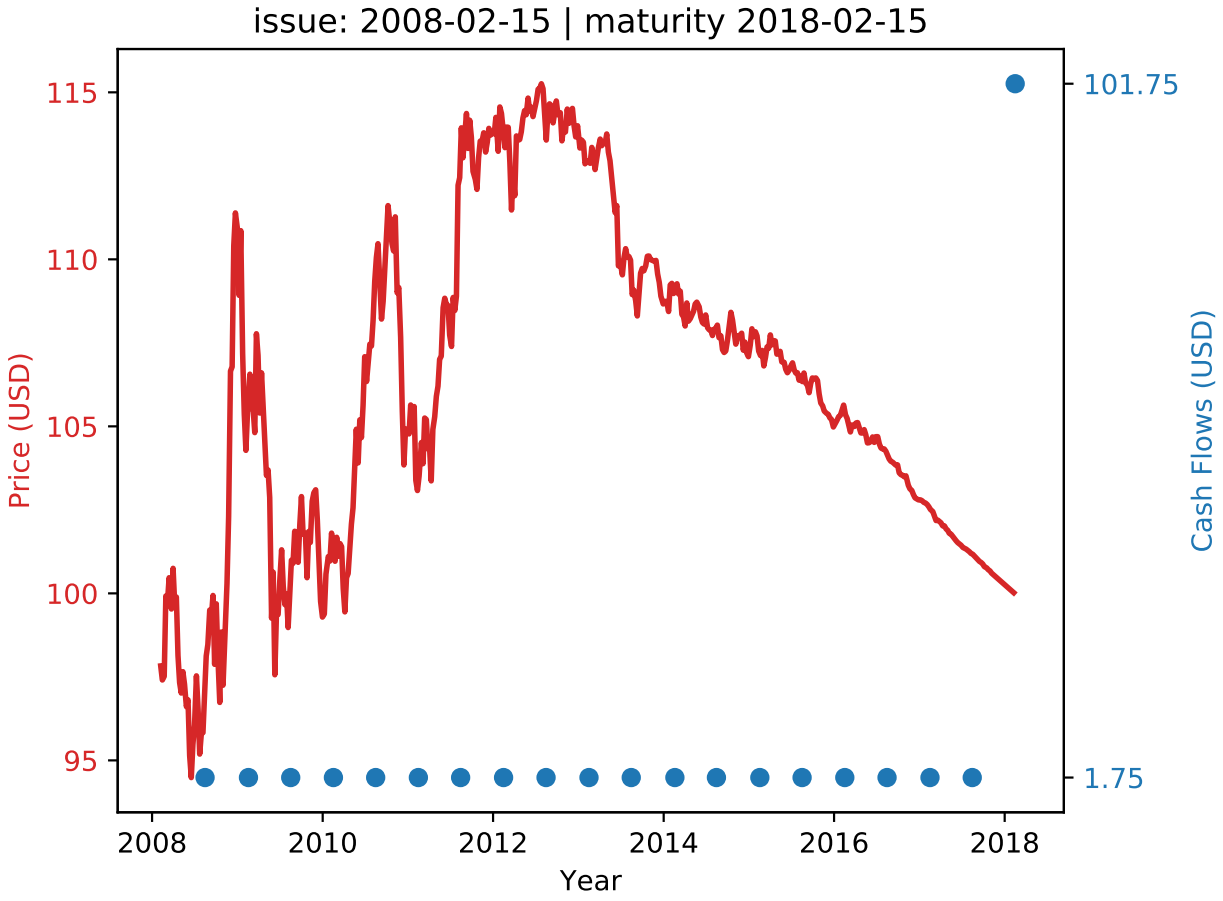
TREASURY AUCTION RESULTS

Term and Type of Security	30-Year Bond
CUSIP Number	912810SP4
Series	Bonds of August 2050
Interest Rate	1-3/8%
High Yield [1]	1.406%
Allotted at High	20.72%
Price	99.243433
Accrued Interest per $1,000	$0.07473
Median Yield [2]	1.320%
Low Yield [3]	1.150%
Issue Date	August 17, 2020
Maturity Date	August 15, 2050
Original Issue Date	August 17, 2020
Dated Date	August 15, 2020

	Tendered	Accepted
Competitive	$55,533,455,000	$25,991,895,000
Noncompetitive	$8,118,800	$8,118,800
FIMA (Noncompetitive)	$0	$0
Subtotal [4]	**$55,541,573,800**	**$26,000,013,800[5]**
SOMA	$15,463,573,000	$15,463,573,000
Total	**$71,005,146,800**	**$41,463,586,800**

	Tendered	Accepted
Primary Dealer [6]	$34,323,000,000	$7,354,440,000
Direct Bidder [7]	$5,236,400,000	$3,086,400,000

The Life a Treasury Note



Treasury Checklist

- ✓ Yields and coupon rates are quoted as APRs, so divide them by the number of payments per year to get the per-period rate
- ✓ Multiply the number of years by the number of payments per year to get the number of periods
- ✓ The future value is the face value
- ✓ Remember: if the coupon rate is greater than (less than) the yield, the price is greater than (less than) the face value
- ✓ **Use four decimal places!**

Example

On 2012-01-31, the yield on a 100 USD 5.0-year Treasury that matured on 2015-01-31 was 0.3035%. It paid a 2.25% coupon. What should it have traded for? (Hint: the number of payments up to and including maturity is 6.0)

Example

On 2012-01-31, the yield on a 100 USD 5.0-year Treasury that matured on 2015-01-31 was 0.3035%. It paid a 2.25% coupon. What should it have traded for? (Hint: the number of payments up to and including maturity is 6.0)

$$
\begin{aligned}
P_0 \overset{\text{should}}{=} & -\mathrm{PV}(\\
& \quad \mathrm{N} = 6.0, \\
& \quad \mathrm{I/Y} = 0.3035\%/2, \\
& \quad \mathrm{PMT} = (2.25\% \times 100)/2, \\
& \quad \mathrm{FV} = 100) \\
= & \ 105.8086 \text{ USD}
\end{aligned}
$$

Example

On 2012-07-31, a 5.0-year Treasury that matured on 2013-07-31 traded for 103.1758 USD. It paid a 3.375% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 2.0)

Example

On 2012-07-31, a 5.0-year Treasury that matured on 2013-07-31 traded for 103.1758 USD. It paid a 3.375% coupon. What was the yield? (Hint: the number of payments up to and including maturity is 2.0)

$$
\begin{aligned}
y_0 \overset{\text{should}}{=} & \; 2 \times \text{RATE}(\\
& \quad \text{N} = 2.0, \\
& \quad \text{PV} = -103.1758, \\
& \quad \text{PMT} = (3.375\% \times 100)/2, \\
& \quad \text{FV} = 100) \\
= & \; 0.1946\%
\end{aligned}
$$

Bond Arbitrage

Arbitrage

- On 2020-01-28, the yield on a one year T-bill was 1.53%.
- What's the fair price?

$$\text{price} = \frac{100 \text{ USD}}{1 + .0153} = 98.4931 \text{ USD}$$

- Because T-bills are risk-free, we ignore margin requirements
- Assume that you can borrow from and lend to a counterparty (call it a "bank") for a year at 1.53%

Arbitrage

- What if the T-bill is trading for 97.5082 USD?
- Cash Flow Table

	2020-01-28	**2021-01-28**
Buy the T-bill	-97.5082 USD	+100 USD
Bank	+98.4931 USD	-100 USD
Net Cash Flow	+.9849 USD	0 USD

- Borrow 98.4931 USD, use 97.5082 USD of it to buy the T-bill
- In one year's time, you'll owe the bank 100 USD but the T-bill will pay you 100 USD

Arbitrage

- What if the T-bill is trading for 99.4780 USD?
- Cash Flow Table

	2020-01-28	**2021-01-28**
Sell the Bond	+99.4780 USD	-100 USD
Deposit the Proceeds	-98.4931 USD	+100 USD
Net Cash Flow	+.9849 USD	0 USD

- Your friend Eva at the bank owns the T-bill

  You: Hey Eva, can I borrow your T-bill?
  Eva: You bet. Just pay me 100 USD in one year

The Yield Curve

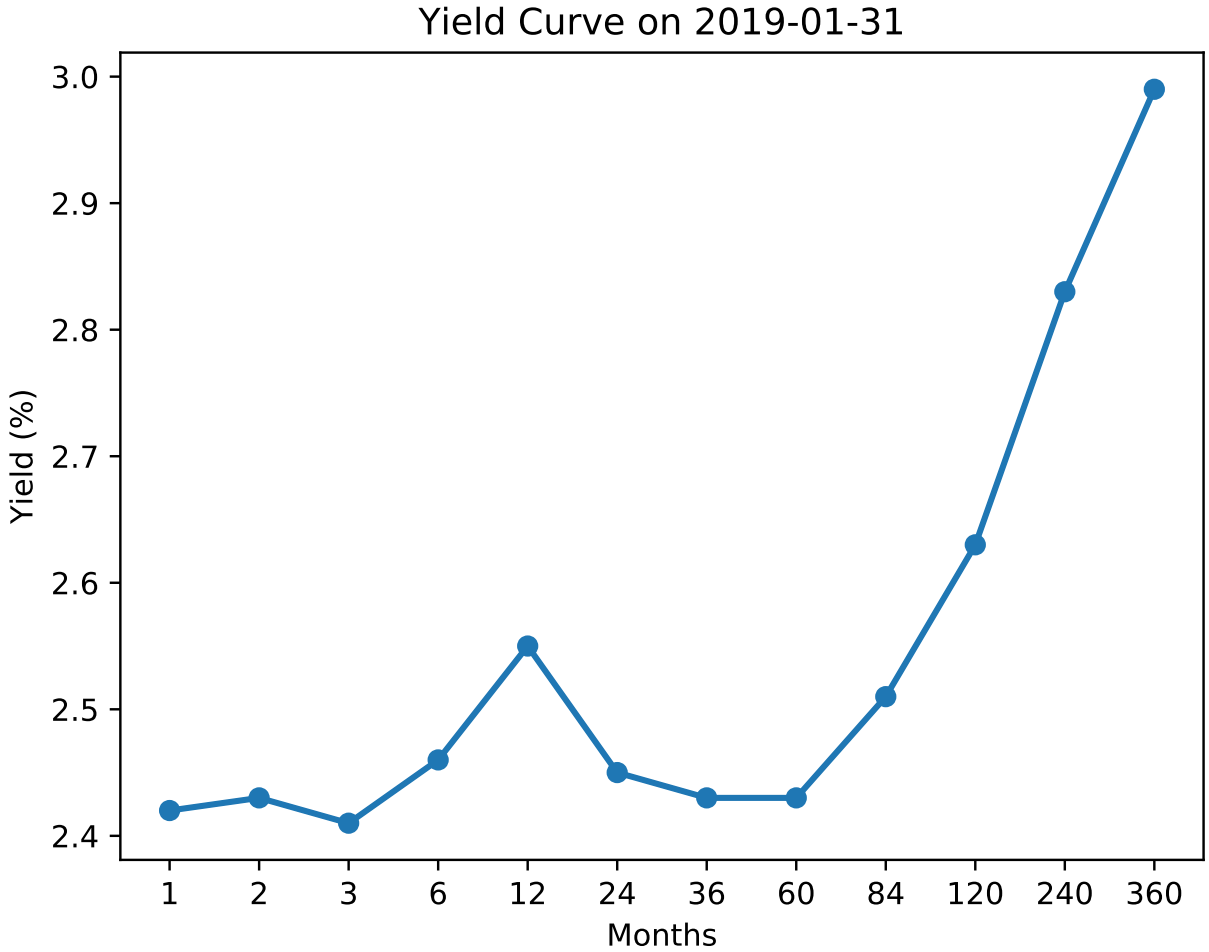
The Yield Curve

Yield Curve on 2020-08-28

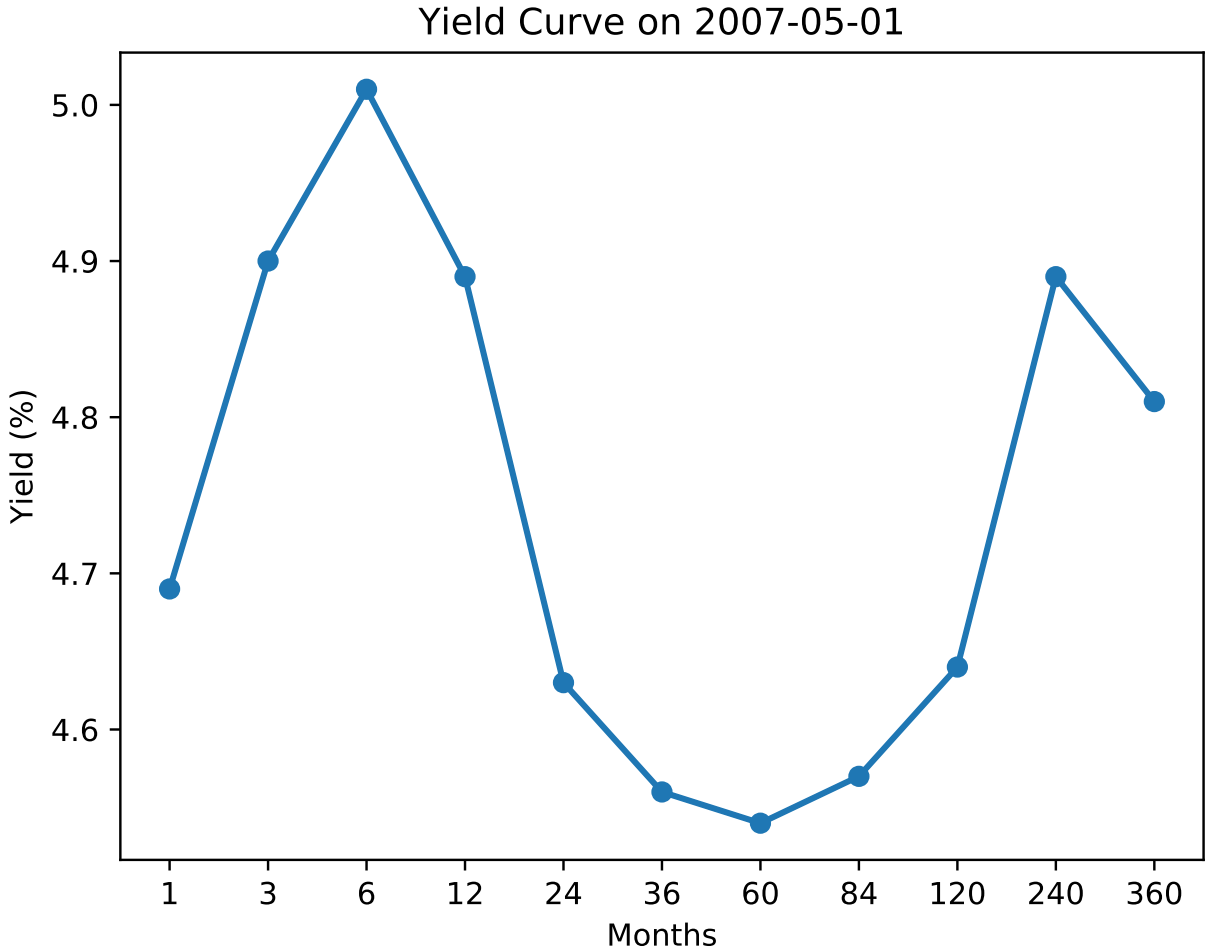
6mo	1yr	2yr	3yr	5yr	7yr	10yr	20yr	30yr
0.11	0.12	0.14	0.16	0.28	0.51	0.74	1.29	1.52

- The yield curve can be found in many places, but the best place is directly from the Treasury (`treasury.gov/data`)
- The yield is the yield on a bond trades at par
- For example, the 30-year yield is the yield-to-maturity of a 100 USD 30-year T-bond that trades for 100 USD

A typical yield curve



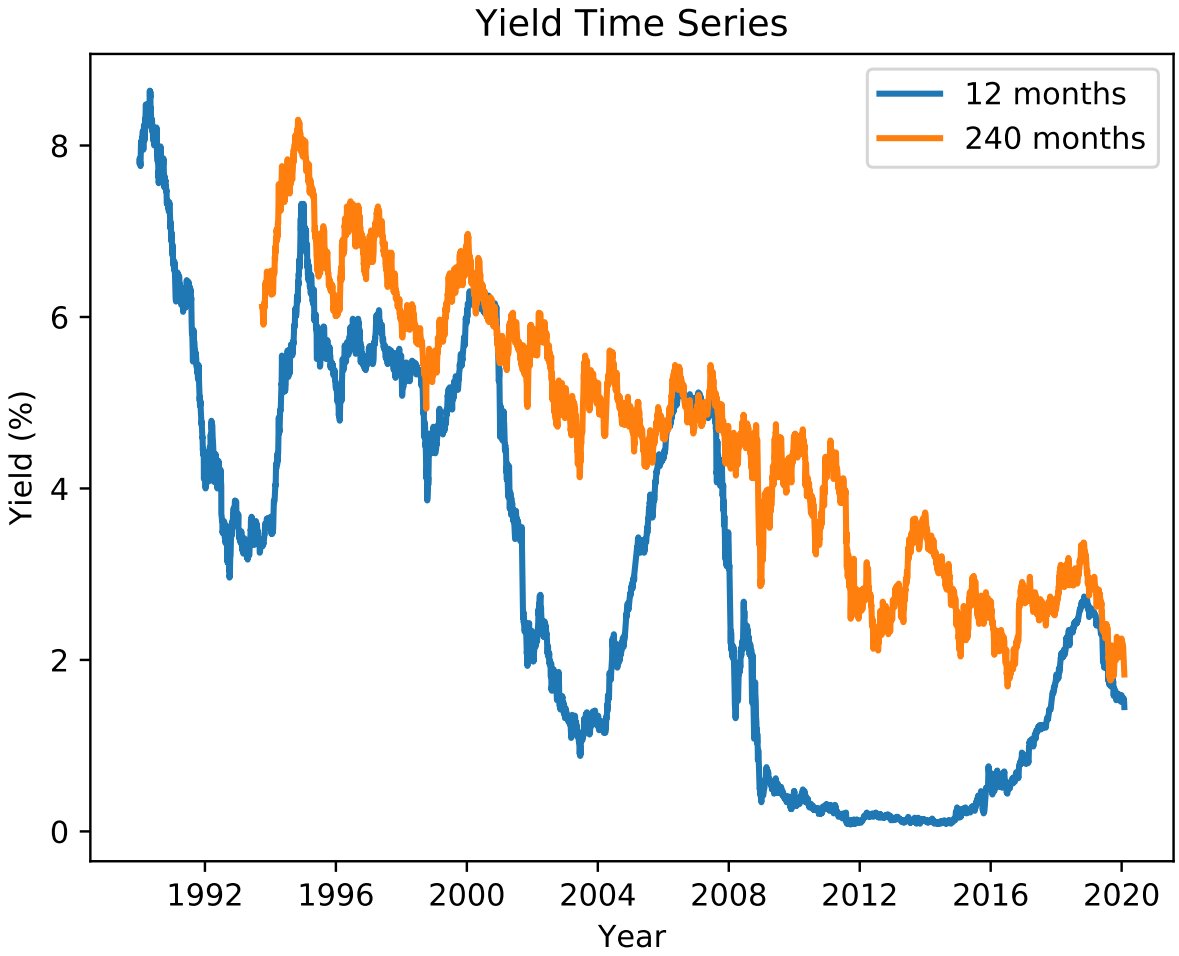
Entering a recession



Why are there different rates?

- Why are there different rates? Why isn't there just one rate for all terms?
- When conditions are expected to improve, the curve is upward sloping; when they are expected to deteriorate, the curve is downward sloping
- Usually, the curve is upward sloping
- When long-term rates (e.g. 20 year) fall below short-term rates (e.g. 1 year), we say that the yield curve has <u>inverted</u>

In the time-series...



Zero and Forward Rates

Zero Rates

- We can use the yield curve to compute the rates on zero-coupon bonds of various terms called zero rates through a process known as bootstrapping
- y_1, y_2 are the 6-and 12-month yields from the yield curve
- z_1, z_2 are the 6-and 12-month zero rates
- The 6-month zero rate is the 6-month yield from the yield curve: $z_1 = y_1$
- Solve for z_2:

$$P = F = \frac{y_2 F/2}{1 + z_1/2} + \frac{y_2 F/2}{(1 + z_2/2)^2} + \frac{F}{(1 + z_2/2)^2}$$

- This equation says that z_2 is the zero rate for a 12-month, semi-annual coupon bond that trades at par

Zero Rates

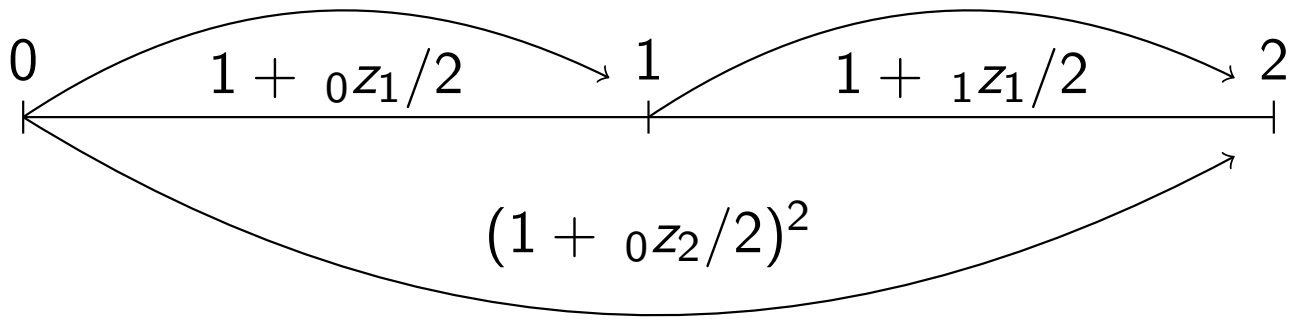
Suppose that $y_1 = 1\%$ and $y_2 = 3\%$. What are the zero rates?

Zero Rates

Suppose that $y_1 = 1\%$ and $y_2 = 3\%$. What are the zero rates?
$z_1 = y_1 = 1\%$. Use the following equation for z_2

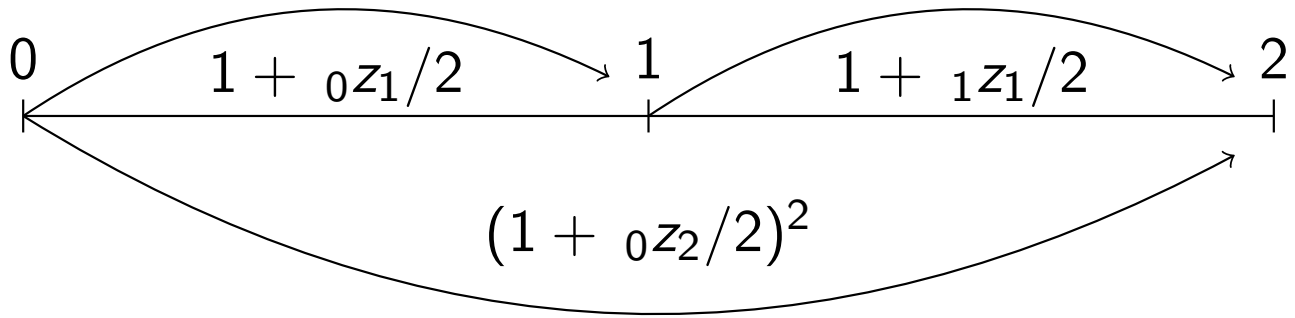
$$
\begin{aligned}
z_2 = 2 \times \mathrm{RATE}(& \\
& \mathrm{N} = 2, \\
& \mathrm{PV} = -(1 - (y_2/2)/(1 + z_1/2)), \\
& \mathrm{PMT} = 0, \\
& \mathrm{FV} = y_2/2 + 1) \\
= 3.0152\% &
\end{aligned}
$$

Forward Rates



- ${}_0z_1$ is the yield today (date 0) on a 6-month zero coupon bond that matures in six months (date 1)
- ${}_0z_2$ is the yield today (date 0) on a 12-month zero coupon bond that matures in one year (date 2)
- ${}_1z_1$ is the yield in six months (date 1) on a 6-month zero coupon bond that matures in one year (date 2)

Forward Rates



$$(1 + {}_0z_1/2)(1 + {}_1z_1/2) \overset{\text{should}}{=} (1 + {}_0z_2/2)^2 \tag{1}$$

- The current yield curve $({}_0z_1, {}_0z_2)$ predicts the future short-term yield $({}_1z_1)$, which we call the <u>forward rate</u>
- This relationship between current yields and future yields is called the <u>expectations hypothesis</u>

Forward Rates

${}_0z_1 = 1.57\%$ and ${}_0z_2 = 1.58\%$. What did the market expect the six-month yield to be in six months (${}_1z_1$)?

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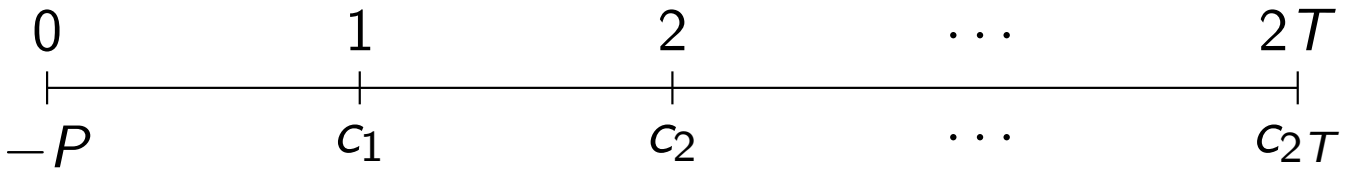
$$ {}_1z_1 = 2\left(\frac{(1 + {}_0z_2/2)^2}{(1 + {}_0z_1/2)} - 1\right) = 1.5879\% \tag{2} $$

Prices and Yields Revisited

Using the Yield Curve to Price Bonds

Consider a T-year stream of semi-annual cash-flows.



where c_1 is the six-month (0.5 year) cash-flow, c_2 twelve-month (1 year), c_3 eighteen-month (1.5 year), and so on

$$P \overset{\text{should}}{=} \frac{c_1}{(1+z_1/2)} + \frac{c_2}{(1+z_2/2)^2} + \frac{c_3}{(1+z_3/2)^3} + \cdots + \frac{c_{2T}}{(1+z_{2T}/2)^{2T}}$$

where z_1 is the six-month (0.5 year) zero-coupon yield (APR), z_2 twelve-month (1 year), z_3 eighteen-month (1.5 year), and so on

Using the Yield Curve to Price Bonds

What is the **price** of a 2-year, 1000 USD 20% coupon bond that makes semi-annual coupon payments. Suppose that the 6, 12, 18, and 24 month zero rates are 1%, 3%, 4%, and 4.5% respectively. What should the 2-year bond trade for?

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First, compute the present values:

$$\text{pv}_1 = \frac{100 \text{ USD}}{(1 + 0.010/2)^1} = 99.5025 \text{ USD} \tag{3}$$

$$\text{pv}_2 = \frac{100 \text{ USD}}{(1 + 0.030/2)^2} = 97.0662 \text{ USD} \tag{4}$$

$$\text{pv}_3 = \frac{100 \text{ USD}}{(1 + 0.040/2)^3} = 94.2322 \text{ USD} \tag{5}$$

$$\text{pv}_4 = \frac{100 \text{ USD}}{(1 + 0.045/2)^4} = 91.4843 \text{ USD} \tag{6}$$

Next, note that the cash-flows from the zero coupon bond are 100 USD, 100 USD, 100 USD, and 1100 USD.

$$\text{pv}_{\text{cpn}} = 1 \times \text{pv}_1 + 1 \times \text{pv}_2 + 1 \times \text{pv}_3 + 11 \times \text{pv}_4 = 1297.1286 \text{ USD} \tag{7}$$

Yield-to-Maturity (YTM)

- Rather than carry around an entire yield curve whenever we want to recompute the price of a bond, we summarize the information with the yield-to-maturity (YTM), with which you are familiar from your previous coursework
- Example: consider a 1000 USD, 30-year, 5% T-bond that makes semi-annual payments, and trades for 1011.9610 USD
- The YTM on this T-bond is the one rate that would make the PV of cash-flows equal to the price, namely, 4.9233%

Yield-to-Maturity (YTM)

What is the **yield** of a 2-year, 1000 USD 20% coupon bond that makes semi-annual coupon payments. Suppose that the 6, 12, 18, and 24 month zero rates are 1%, 3%, 4%, and 4.5% respectively. What should the 2-year bond trade for?

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From the previous problem,

$$
\begin{aligned}
y_0 = 2 \times \mathrm{RATE}(& \\
& \mathrm{N} = 4, \\
& \mathrm{PV} = -1297.1286, \\
& \mathrm{PMT} = (20\% \times 100)/2, \\
& \mathrm{FV} = 1000) \\
= 4.3307\%
\end{aligned}
$$